ML Day 7

23 April 2026 06:02 PM

Vectorization in NLP

→ Process of converting text data into numerical format

1) Bag of words (BOW)

→ Count how many times each word appears in the document

→ Ignore grammar

4

S1 → "I love NLP"
S2 → "I love AI"
} → ["I", "Love", "NLP", "AI"]

Sentence	I	love	NLP	AI
S1	1	1	1	0
S2	1	1	0	1

Doc

→ Importance

2) TFIDF [TF − IDF]

TF → Term Frequency

IDF → Inverse Document frequency

Convert text into Numerical values by giving imp to words in a document

TF → measures how frequently a words appears in a 1 document

$$TF(t,d) = \frac{\text{No. of times term t appears in a document}}{\text{Total no. of terms in document}}$$

more → Higher TF

If word appears more → Higher TF

IDF → Inverse Document frequency
→ measures how imp a word is across all documents.

$$IDF(t) = \log\left(\frac{N}{1 + DF(t)}\right)$$

N → total no. of documents
DF(t) → no. of doc contains the word

Email

1) Name, email
2) upload dataset → machine → NR
3) enter → filter → meet

text
spam/notspam
1 → Spam
1
0 → N.S
0
1

Data → LR → NLP → New data → o/p

Projects:(Compulsory should post it on LinkedIn by tagging everyone)

1. **Sentiment Analysis App (Movie/Product Reviews)**
--> review: positive or negative

2. **Fake News Detection System**
--> Fake / Real

3. **Resume Classification System**

-- After uploading resume, automatically classify into categories, whether it is Data science, Web development, Marketing

4. Customer Support Ticket Classifier
-- classify customer queries into :
Billing, technical issues, general query

5. Toxic comment detection
-- dataset -->social media comments dataset
-- detect whether it is Toxic/safe

Tag:
Manohar Chary V
Raghu ram aduri
Vishwanath nyathani
Kalpana katiki reddy
Innomatics research labs